

RAMANJANEYULU GANIPISETTY

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Summary

Mechatronics Engineering MSc student with experience in robotics, simulation-based system development, data analysis, and engineering validation. Proficient in Python, NumPy, Pandas, MATLAB, and Webots, with experience applying analytical and problem-solving techniques to autonomous systems and performance evaluation. Strong interest in machine learning, robotics, optimisation, and data-driven decision making, with a proven ability to analyse complex systems, interpret performance data, and deliver practical solutions within fast-paced environments. Seeking to apply technical, analytical, and research skills within an innovative Applied Science team.

Employment History

AMAZON NCL-1

AFE Team Lead

Gateshead

November 2023 to November 2025

- Lead a team of 70+ associates, ensuring adherence to safety and operational standards while minimizing incidents through training and monitoring.
- Analysed operational performance datasets and key metrics to identify trends, support process optimisation, and improve workflow efficiency across high-volume operations.
- Managed complex documentation systems and data analytics tools to track performance metrics, ensuring 100% adherence to operational and safety standards.
- Coached a large team on strictly following Standard Operating Procedures (SOPs) and maintaining a safe, organized '5S' environment.
- Adapt to dynamic operational environments by managing fluctuating workloads and shifting priorities, meeting increased processing demands of 75,000+ shipments during peak periods.

Engineering Projects

Aircraft Landing Gear

University project | 2022 - 2023

- Designed an innovative aircraft landing gear strut incorporating an auxetic lattice core sandwiched between composite material layers to improve energy absorption and structural performance.
- Conducted Finite Element Analysis (FEA) to evaluate stress distribution, deformation characteristics, and load-bearing capability under simulated landing conditions.
- Applied engineering principles, material selection techniques, and structural optimisation methods to enhance impact resistance while reducing component weight.
- Produced detailed engineering documentation and analysed simulation results to support design validation and performance assessment.

Gastrointestinal Diagnostic Robot Development

University project | 2026 - Present

- Developing autonomous navigation algorithms for a robotic diagnostic platform using simulation-based testing and performance evaluation.
- Utilising Webots simulation software to model robotic behaviour, sensor integration, navigation strategies, and system performance in a virtual environment.
- Applying robotics, embedded systems, and control engineering principles to support autonomous movement and data acquisition within constrained environments.
- Analysing system performance data to evaluate navigation accuracy and optimise control strategies.

Education

University of Chester – MSc
Mechanical engineering and Mechatronics
Chester, Cheshire, England

Jan 2026 to Jan 2028

- Expected outcome – Distinction
- Course curriculum – Applied finite element analysis, Robotics, Advanced embedded systems and internet of things, advanced manufacturing process and systems.

Newcastle College - BSc (Hons)
Mechanical Manufacturing Engineering

Sep 2021 to Sep 2023

Newcastle upon Tyne, England

- Grade Percentage - Merit
- Course Curriculum - Finite Element Analysis, Computer-aided Modelling and analysis, Advanced mechanical science, Project Management, Manufacturing systems and processes, Material Science, Thermal Engineering, Dissertation.

Courses

Introduction to Robotics – IIT Madras
Issued by NPTEL

November 2025 to January 2026

- Applied robotics principles to analyse robotic motion, kinematic relationships, and control requirements in automated systems..
- Familiarity with robotic manipulator design and operation, including motion planning, workspace analysis, and automation fundamentals.

Python - IIT Madras

Issued by NPTEL

November 2023 to January 2024

- Applied programming techniques using Python for developing structured, maintainable, and efficient solutions to computational challenges.
- Familiarity with core data structures and algorithm design principles, enabling effective problem decomposition and optimization.

Skills

1. Technical Skills:

- Python (NumPy, Pandas)
- MATLAB
- Webots
- Autonomous Systems
- Sensor Integration
- Embedded systems

2. Engineering Skills

- Experimental Design
- Technical Reporting
- Prototype development
- Root cause analysis
- Validation Testing

3. Process improvement

- Team leadership
- Stakeholder communication
- Safety compliance
- Problem solving
- Project coordination